JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON THURSDAY 02nd APRIL 2026) **TIME : 9:00 AM TO 12:00 NOON**

MATHEMATICS — TEST PAPER WITH SOLUTION

SECTION-A

1. Let $\alpha, \alpha + 2, \alpha \in \mathbf{Z}$, be the roots of the quadratic equation $x(x + 2) + (x + 1)(x + 3) + (x + 2)(x + 4) + \ldots.. + (x + n - 1)(x + n + 1) = 4n$ for some $n \in N$.

Then $(n + \alpha)$ is equal to :

(1) 0 (2) 1

(3) 2 (4) 4

Sol. $nx^2 + x(2 + 4 + 6 + \ldots.. + 2n) + (1.3 + \ldots.. + (n-1)(n+1)) = 4n$

$$nx^2 + n(n+1)x + \frac{n(n-1)(2n+5)}{6} = 4n$$

$$x^2 + (n+1)x + \frac{(n-1)(2n+5)}{6} = 4$$

D must be a perfect square

$$D = \frac{122 - 2n^2}{6} = 20 - \left(\frac{n^2-1}{3}\right)$$

If make perfect square $\Rightarrow \frac{n^2-1}{3} = 16 \Rightarrow n = 7$

So Equation is

$$\Rightarrow x^2 + 8x + \frac{8 \times 15}{6} - 5 = 0$$

$x^2 + 8x + 15 = 0$

$x = -3, -5$

$\alpha = -5, \ \alpha + 2 = -3$

$\alpha + n = 7 - 5 = 2$

2. Let x and y be real number such that $50\left(\frac{2x}{1+3i} - \frac{y}{1-2i}\right) = 31 + 17i$, $i = \sqrt{-1}$. Then the value of $10(x - 3y)$ is :

(1) 20 (2) 31

(3) 35 (4) 75

Sol. $50\left(\frac{2x(1-3i)}{10} - \frac{y(1+2i)}{5}\right) = 31 + 17i$

$10x - (30x)i - 10y - 20y\,i = 31 + 17i$

$$\left.\begin{aligned} 10(x-y) &= 31 \\ -30x - 20y &= 17 \end{aligned}\right\}$$

$30x - 30y = 93$

$-10(3x + 2y) = 17$

$x = 0.9, \ y = -2.2$

$10(x - 3y) = 75$

3. Let $\alpha, \beta \in \mathbf{R}$ be such that the system of linear equations

$x + 2y + z = 5$

$2x + y + \alpha z = 5$

$8x + 4y + \beta z = 18$

has no solution. Then $\frac{\beta}{\alpha}$ is equal to :

(1) –4 (2) 4

(3) 8 (4) –8

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Sol. $x + 2y + z = 5$

$2x + y + \alpha z = 5$

$8x + 4y + \beta z = 18$

for no solution

$$\Delta = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & \alpha \\ 8 & 4 & \beta \end{vmatrix} = 0$$

$\Delta = 1(\beta - 4\alpha) - 2(2\beta - 8\alpha) + 1\,(0) = 0$

$= \beta - 4\alpha - 4\beta + 16\alpha = 0$

$\Rightarrow 4\alpha = \beta$

4. Let $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$.

If $A^2 - 4A + I = O$ and $B^2 - 5B - 6I = O$, then among the two statements :

(S1) : $[(B - A)(B + A)]^T = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$

and

(S2) : det (adj(A + B)) = – 5,

(1) only (S1) is correct

(2) only (S2) is correct

(3) both (S1) and (S2) are correct

(4) both (S1) and (S2) are wrong

Sol. $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$

$A^2 - 4A + I = 0$

trace of matrix

$\alpha + 1 = 4$

$\alpha = 3$

$B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$

$B^2 - 5B - 6I = 0$

det. (B) = $6 - 3\beta = -6$

$3\beta = 12$

$\beta = 4$

$B - A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}; B + A = \begin{bmatrix} 4 & 5 \\ 5 & 5 \end{bmatrix}$

$(B - A)(B + A) = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$

$\text{adj}(A) = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$, $\text{adj}B = \begin{bmatrix} 2 & -3 \\ -4 & 3 \end{bmatrix}$

$\text{adj}(A + B) = \begin{bmatrix} 5 & -5 \\ -5 & 4 \end{bmatrix}$

5. Let A be the set of first 101 terms of an A.P., whose first term is 1 and the common difference is 5 and let B be the set of first 71 terms of an A.P., whose first term is 9 and the common difference is 7. Then the number of elements is $A \cap B$, which are divisible by 3, is :

(1) 4 (2) 5

(3) 6 (4) 7

Sol. AP_1 Set A = {1, 6,11,16 ..... 101 terms}

AP_2 Set B = {9, 16 ..... 71 terms}

D = L.C.M $\{d_1, d_2\} = 35$

1st Common term is 16

$16 + (n - 1)\,35 \leq 499$

$n \leq 14.8$

$\Rightarrow n = 14$

$A \cap B$ = {16,51,86,121,156, 191, 226, 261, 296, 331, 366,401,436,471}

terms divisible by '3'

= {51,156,261,366,471}

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6. The number of seven-digit numbers, that can be formed by using the digits 1, 2, 3, 5 and 7 such that each digit is used at least once, is ;

(1) 15400 (2) 17800

(3) 16800 (4) 29400

Sol. Exactly one digit repeated $\rightarrow {}^5C_1 \times \frac{7!}{3!} = 4200$

Exactly two digits repeated $\rightarrow {}^5C_2 \times \frac{7!}{2!2!} = 12600$

Total numbers = 16800

7. The number of elements in the set

$S = \left\{(r, k) : k \in \mathbf{Z} \text{ and } {}^{36}C_{r+1} = \frac{6({}^{35}C_r)}{(k^2-3)}\right\}$, is

(1) 2 (2) 4 (3) 8 (4) 16

Sol. $\frac{36}{r+1}.{}^{35}C_r = 6.\frac{{}^{35}C_r}{k^2-3}$

$k^2 - 3 = \frac{r+1}{6}$

$k^2 = \frac{r+19}{6}$

$0 \le r \le 35$

$19 \le r + 19 \le 54$

$3.1 \le \frac{r+19}{6} \le 9$

$\therefore k^2 = 4, 9$

$k = \pm 2, \pm 3$

$\therefore (r, k) \Rightarrow (5, \pm 2) \Rightarrow (35, \pm 3)$

$\therefore \quad 4$

8. If the mean of the data

Class	5-10	10-15	15-20	20-25	25-30	30-35
Frequency	2	k	28	54	k+1	5

is 21, then k is one of the roots of the equation :

(1) $2x^2 - 23x - 10 = 0$

(2) $4x^2 - 35x + 24 = 0$

(3) $2x^2 - 19x - 10 = 0$

(4) $2x^2 - 35x + 98 = 0$

Sol.

Class	5-10	10-15	15-20	20-25	25-30	30-35
Freq.	2	k	28	54	k+1	5

mean

$21 = \frac{15 + 12.5k + 490 + 1215 + 27.5 + 27.5k + 162.5}{90 + 2k}$

$1890 + 42k = 1910 + 40k$

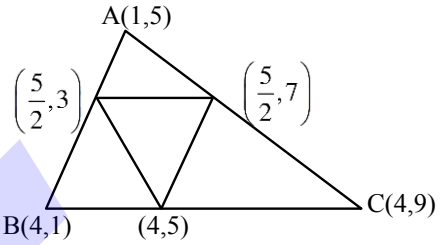
$k = 10$

9. Let the mid points of the sides of a triangle ABC be $\left(\frac{5}{2}, 7\right)$, $\left(\frac{5}{2}, 3\right)$ and (4, 5). If its incentre is (h, k), then 3h + k is equal to :

(1) 11 (2) 12

(3) 13 (4) 14

Sol.



$I = \left(\frac{24}{9}, 5\right)$

$\therefore 3h + k = 8 + 5 = 13$

10. Let an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, a < b, pass through the point (4, 3) and have eccentricity $\frac{\sqrt{5}}{3}$. Then the length of its latus rectum is :

(1) $\frac{4\sqrt{5}}{3}$ (2) $2\sqrt{5}$

(3) $\frac{7\sqrt{5}}{3}$ (4) $\frac{8\sqrt{5}}{3}$

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Sol. $e^2 = 1 - \frac{a^2}{b^2} \Rightarrow \frac{a^2}{b^2} = 1 - e^2 = 1 - \frac{5}{9} = \frac{4}{9}$

$\Rightarrow \frac{a^2}{b^2} = \frac{4}{9}$ …(1)

Passes through (4,3) $\Rightarrow \frac{16}{a^2} + \frac{9}{b^2} = 1$ …(2)

from (1) and (2)

$a^2 = 20$ and $b^2 = 45$

$LR = \frac{2a^2}{b} = \frac{2(20)}{3\sqrt{5}} = \frac{8\sqrt{5}}{3}$

11. If $\sin\left(\frac{\pi}{18}\right)\sin\left(\frac{5\pi}{18}\right)\sin\left(\frac{7\pi}{18}\right) = K$, then the value of $\sin\left(\frac{10K\pi}{3}\right)$ is :

(1) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (2) $\frac{\sqrt{3}-1}{\sqrt{2}}$

(3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Sol. $K = \sin10° \sin50° \sin70°$

$= \frac{1}{4}\sin30° = \frac{1}{8}$

$\sin10K\frac{\pi}{3} = \sin\left(10 \times \frac{1}{8}.\frac{\pi}{3}\right) = \sin\frac{5\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}}$

12. Let $S = \{x \in [-\pi, \pi] : \sin x (\sin x + \cos x) = a, a \in \mathbf{Z}\}$. Then n(S) is equal to :

(1) 3 (2) 6

(3) 7 (4) 9

Sol. $\sin x (\sin x + \cos x) \in \left[\frac{1-\sqrt{2}}{2}, \frac{1+\sqrt{2}}{2}\right]$

2 integer will be there $\Rightarrow a = 0, 1$

If $a = 0$ $\sin x(\sin x + \cos x) = 0$

$\Rightarrow \sin x = 0$ or $\sin x + \cos x = 0$

$\boxed{x = -\pi, 0, \pi}$ $= \tan x = -1$

3 solutions $\boxed{x = -\frac{\pi}{4}, \frac{3\pi}{4}}$ 2 solutions

If $a = 1$ $\sin(\sin x + \cos x) = 1$

$\sin^2 x + \sin x \cos x = 1$

$2\sin^2 x + 2\sin x \cos x = 2$

$1 - \cos2x + \sin2x = 2$

$\text{Sin}2x - \cos2x = 1 \rightarrow$ square

$\text{Sin } 4x = 0$

$4x = -4\pi, -3\pi, \ldots\ldots\ldots, 3\pi, 4\pi$

$x = -\pi, -\frac{3\pi}{4}, -\frac{\pi}{2}, -\frac{\pi}{4}, 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$

$\Rightarrow x = -\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{4}, \frac{\pi}{2}$ 4 solutions

Total 9 solution

13. If the point of intersection of the lines $\frac{x+1}{3} = \frac{y+a}{5} = \frac{z+b+1}{7}$ and $\frac{x-2}{1} = \frac{y-b}{4} = \frac{z-2a}{7}$ lies on xy-plane, then the value of a + b is :

(1) 2 (2) 5

(3) 7 (4) 9

Sol. Line (L_1)

$\frac{x+1}{3} = \frac{y+a}{5} = \frac{z+b+1}{7} = r_1$

$\Rightarrow$ General point P on L_1 is

$(3r_1 - 1, 5r_1 - a, 7r_1 - b - 1)$

Line (L_2)

$\frac{x-2}{1} = \frac{y-b}{4} = \frac{z-2a}{7} = r_2$

$\Rightarrow$ General point Q on L_2 is

$(r_2 + 2, 4r_2 + b, 7r_2 + 2a)$

For point of intersection

$3r_1 - 1 = r_2 + 2 \Rightarrow r_2 = 3r_1 - 3$ …(1)

$5r_1 - a = 4r_2 + b$ …(2)

$7r_1 - b - 1 = 7r_2 + 2a$ …(3)

Since point lies on XY plane

$\Rightarrow$ z – coordinate = 0

From L_1 : $7r_1 - b - 1 = 0 \Rightarrow 7r_1 = b + 1$

From L_2 : $7r_2 + 2a = 0 \Rightarrow 2a = -7r_2$

Substitute $r_2 = 3r_1 - 3$

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$2a = -7(3r_1 - 3)$

$\Rightarrow a = \frac{-21r_1 + 21}{2}$

Put in equation (2)

$5r_1 - a = 4r_2 + b$

$5r_1 - \left(\frac{-21r_1 + 21}{2}\right) = 4(3r_1 - 3) + (7r_1 - 1)$

$\Rightarrow r_1 = \frac{5}{7}$

$\Rightarrow b = 7r_1 - 1 = 4$

$\Rightarrow a = 3 \therefore a + b = 7$

14. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 2$ and $|\vec{b}| = 3$, then the maximum value of $3\left|(3\vec{a} + 2\vec{b})\right| + 4\left|(3\vec{a} - 2\vec{b})\right|$ is :

(1) 30 (2) 36

(3) 60 (4) 72

Sol. $E = 3\sqrt{9a^2 + 4b^2 + 12\vec{a}.\vec{b}} + 4\sqrt{9a^2 + 4b^2 - 12\vec{a}.\vec{b}}$

$= 3\sqrt{36 + 36 + 12 \times 6\cos\theta} + 4\sqrt{36 + 36 - 72\cos\theta}$

$= 3\sqrt{72 + 72\cos\theta} + 4\sqrt{72 - 72\cos\theta}$

$= 18\sqrt{2}\sqrt{1 + \cos\theta} + 24\sqrt{2}\sqrt{1 - \cos\theta}$

$= 18\sqrt{2}\sqrt{2}\cos\frac{\theta}{2} + 24\sqrt{2}\left(\sqrt{2}\sin\frac{\theta}{2}\right)$

$= 36\left(\cos\frac{\theta}{2}\right) + 48\sin\left(\frac{\theta}{2}\right)$

$E_{max} = \sqrt{(36)^2 + (48)^2} = \sqrt{6^4 + 6^2 \times 8^2} = 60$

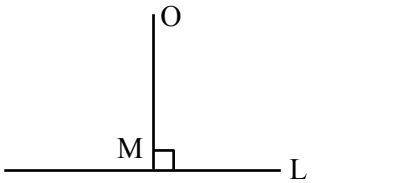
15. Let a line L passing through the point (1, 1, 1) be perpendicular to both the vectors $2\hat{i} + 2\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + \hat{k}$. If P(a, b, c) is the foot of perpendicular from the origin on the line L, then the value of 34(a + b + c) is :

(1) 50 (2) 80

(3) 100 (4) 120

Sol. Direction vector of line is

$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix} = -2\hat{i} + 3\hat{j} - 2\hat{k}$



$\frac{x-1}{2} = \frac{y-1}{-3} = \frac{z-1}{2}$

So equation of line is

$\frac{x-1}{2} = \frac{y-1}{-3} = \frac{z-1}{2} = \lambda$ let

let $(2\lambda + 1, -3\lambda + 1, 2\lambda + 1)$ be the foot of perpendicular from (0, 0, 0) upon given line

$\Rightarrow \overrightarrow{OM}.\left(2\hat{i} - 3j + 2\hat{k}\right) = 0$

$\Rightarrow 2(2\lambda + 1) - 3(-3\lambda + 1) + 2(2\lambda + 1) = 0$

$\Rightarrow \lambda = -\frac{1}{17}$

So $(a, b, c) \equiv \left(\frac{-2}{17} + 1, \frac{3}{17} + 1, \frac{-2}{17} + 1\right)$

$\Rightarrow 34(a + b + c) = 100$

16. If $\lim\limits_{x \to 2} \frac{\sin\left(x^3 - 5x^2 + ax + b\right)}{(\sqrt{x-1} - 1)\log_e(x-1)} = m$, then a + b + m is equal to:

(1) 5 (2) 6

(3) 8 (4) 10

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Sol. $\because$ Denominator = 0 at x = 2

$\Rightarrow$ Numerator = 0 at x = 2

$\Rightarrow 2^3 - 5(2)^2 + a(2) + b = 0$

$\Rightarrow 2a + b = 12 \quad ... (1)$

$$\therefore m = \lim_{x\to 2} \frac{\frac{\sin(x^3-5x^2+ax+b)}{(x^3-5x^2+ax+b)}.(x^3-5x^2+ax+b)}{\left(\frac{x-1-1}{\sqrt{x-1}+1}\right).\frac{\log_e(1+(x-2))}{(x-2)}.(x-2)}$$

$$\Rightarrow m = \lim_{x\to 2} 2\left(\frac{x^3-5x^2+ax+b}{(x-2)^2}\right)$$

$$\Rightarrow m = \lim_{x\to 2} 2\left(\frac{3x^2-10x+a}{2(x-2)}\right)$$

$\because$ Denominator = 0 at x = 2

$\Rightarrow$ Numerator = 0 at x = 2

$\Rightarrow 3(2)^2 - 10(2) + a = 0 \Rightarrow a = 8$

put a = 8 in (1), we get b = – 4

$$\therefore m = \lim_{x\to 2} 2\left(\frac{6x-10}{2}\right) \Rightarrow m = 2$$

$\therefore a + b + m = 6$

17. If the curve y = f(x) passes through the point (1, e) and satisfies the differential equation $dy = y(2 + \log_e x)\,dx$, x > 0, then f(e) is equal to :

(1) e^e (2) e^{e^2}

(3) e^{2e} (4) e^{2^e}

Sol. $$\frac{dy}{dx} = y(2+\ell nx)$$

$$\int \frac{dy}{y} = \int (2+\ell nx)\,dx$$

$$\ell ny = 2x + x\ell nx - x + C$$

$$\ell ny = x + x\ell nx + C$$

Since it passes through (1,e)

$1 = 1 + 0 + C \Rightarrow C = 0$

$\ell ny = x + x\ell nx$

$f(x) = y = e^{x + x\ln x}$

$\Rightarrow f(e) = e^{e+e} = e^{2e}$

18. The number of critical points of the function

$$f(x) = \begin{cases} \left|\frac{\sin x}{x}\right| & , \quad x \neq 0 \\ 1 & , \quad x = 0 \end{cases}$$ in the interval $(-2\pi, 2\pi)$ is equal to:

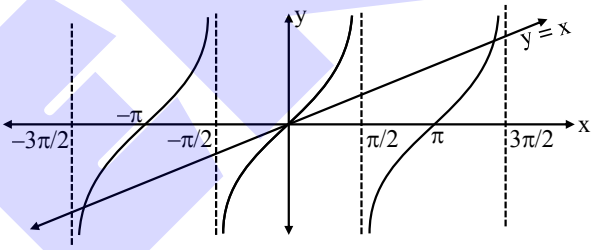
(1) 1 (2) 3

(3) 5 (4) 7

Sol. $\lim_{x\to 0}\left|\frac{\sin x}{x}\right| = 1 = f(0) \rightarrow f(x)$ is continuous

Now $$\frac{d}{dx}\left(\frac{\sin x}{x}\right) = \frac{x\cos x - \sin x}{x^2}$$

$\Rightarrow f'(x) = 0 \Rightarrow \tan x = x$



3 solution in $(-2\pi, 2\pi)$

also f'(0) does not exist at $x = -\pi, \pi$

total 5 points

19. Let [·] denote the greatest integer function. Then the value of $\int_0^3 \left(\frac{e^x + e^{-x}}{[x]!}\right) dx$ is:

(1) $e^2 + e^3 - \frac{1}{e^2} - \frac{1}{e^3}$

(2) $\frac{1}{2}\left(e^2 + e^3 - \frac{1}{e^2} - \frac{1}{e^3}\right)$

(3) $e^2 + e^3 - \frac{1}{2e^2} - \frac{1}{2e^3}$

(4) $\frac{1}{2}(e^2 + e^3) - \frac{1}{e^2} - \frac{1}{e^3}$

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Sol. $\int_0^1 \frac{e^x+e^{-x}}{1}dx + \int_1^2 \frac{e^x+e^{-x}}{1}dx + \int_2^3 \frac{e^x+e^{-x}}{2}dx$

$\int_0^2 e^x dx + \int_0^2 e^{-x}dx + \frac{1}{2}\int_2^3 (e^x+e^{-x})dx$

$(e^2-1) + \frac{(e^{-2}-1)}{-1} + \frac{1}{2}\left((e^3-e^2)-(e^{-3}-e^{-2})\right)$

$\frac{e^3+e^2}{2} + 1 - 1 - \frac{e^{-2}}{2} - \frac{e^{-3}}{2}$

$\frac{e^3+e^2-e^{-2}-e^{-3}}{2}$

20. Let y = y(x) be the solution curve of the differential equation $(1+\sin x)\frac{dy}{dx} + (y+1)\cos x = 0$, $y(0) = 0$. If the curve $y = y(x)$ passes through the point $\left(\alpha, \frac{-1}{2}\right)$, then a value of α is :

(1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$

(3) $\frac{\pi}{3}$ (4) $\frac{\pi}{2}$

Sol. $\int \frac{dy}{y+1} = -\int \frac{\cos x}{1+\sin x}dx$

$\Rightarrow \ell n(y+1) = -\ell n(1+\sin x) + C$

$\because y(0) = 0 \Rightarrow C = 0$

$\Rightarrow y+1 = \frac{1}{1+\sin x}$

passes through $\left(\alpha, -\frac{1}{2}\right)$

$\Rightarrow \sin x = 1$

$\Rightarrow x = \frac{\pi}{2}$ (as per options)

SECTION-B

21. If the domain of the function $f(x) = \sqrt{\log_{(0.6)}\left(\left|\frac{2x-5}{x^2-4}\right|\right)}$ is $(-\infty, a] \cup \{b\} \cup [c,d) \cup (e,\infty)$, then the value of $a+b+c+d+e$ is _____.

Sol. For domain $\log_{0.6}\left|\frac{2x-5}{x^2-4}\right| \geq 0$

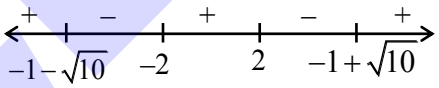
$\left|\frac{2x-5}{x^2-4}\right| \leq 1 \quad \& \quad x \neq \frac{5}{2} \quad \ldots..(1)$

$-1 \leq \frac{2x-5}{x^2-4} \leq 1$

$\frac{2x-5}{x^2-4} + 1 \geq 0$

$\frac{x^2+2x-9}{x^2-4} \geq 0$

$\frac{(x+1)^2-10}{(x-2)(x+2)} \geq 0$



$x \in \left(-\infty, -1-\sqrt{10}\right] \cup (-2,2) \cup \left[-1+\sqrt{10}, \infty\right) \ldots..(2)$

$\frac{2x-5}{x^2-4} - 1 \leq 0$

$\frac{2x-5-x^2+4}{x^2-4} \leq 0$

$\frac{x^2-2x+1}{x^2-4} \geq 0$

$\frac{(x-1)^2}{(x-2)(x+2)} \geq 0$

$x \in (-\infty,-2) \cup (2,\infty) \cup \{1\} \quad \ldots.(3)$

$(1) \cap (2) \cap (3)$

$x \in \left(-\infty, -1-\sqrt{10}\right] \cup \{1\} \cup \left[-1+\sqrt{10}, \frac{5}{2}\right) \cup \left(\frac{5}{2}, \infty\right)$

$a+b+c+d+e = -2+1+5 = 4$

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22. If $\sum_{k=1}^{n} a_k = 6n^3$, then $\sum_{k=1}^{6}\left(\frac{a_{k+1}-a_k}{36}\right)^2$ is equal to ____.

Sol. $a_1 + a_2 + ... + a_n = 6n^3$.

$a_1 + a_2 + ... + a_n + a_{n+1} = 6(n+1)^3$

$a_{n+1} = 6(n+1)^3 - 6n^3$

$= 6\left((n+1)-n\right)\left((n+1)^2 + n^2 + n(n+1)\right)$

$a_{n+1} = 6(1)\left(3n^2+3n+1\right)$

$a_n = 6\left(3(n-1)^2 + 3(n-1) + 1\right)$

$= 6\left(3(n^2 - 2n + 1 + n - 1) + 1\right)$

$= 6\left(3(n^2 - n) + 1\right)$

$a_n = 6\,(3n^2 - 3n + 1)$

$$\sum_{k=1}^{6}\left(\frac{6(3k^2+3k+1)-6(3k^2-3k+1)}{36}\right)^2$$

$$\sum_{k=1}^{6}\left(\frac{36k}{36}\right)^2 = \sum_{k=1}^{6} k^2 = \frac{6\times7\times13}{6} = 91$$

23. Let a, b, c $\in \{1,2,3,4\}$. If the probability, that $ax^2 + 2\sqrt{2}\,bx + c > 0$ for all $x \in \mathbf{R}$, is $\frac{m}{n}$, gcd (m,n) = 1, then m + n is equal to _____.

Sol. $8b^2 - 4ac < 0 \Rightarrow 2b^2 < ac$

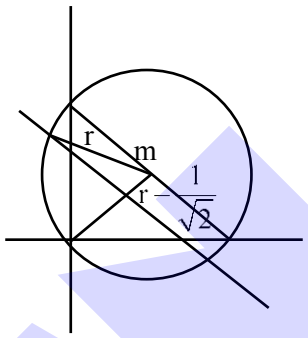
b	a	c		
1	1	3,4	$\rightarrow$	2
1	2	2,3,4	$\rightarrow$	3
1	3	1,2,3,4	$\rightarrow$	4
1	4	1,2,3,4	$\rightarrow$	4
2	3	3,4	$\rightarrow$	2
2	4	3,4	$\rightarrow$	2
				17

Required probability $= \frac{17}{64} \equiv \frac{m}{n}$

$\therefore\ m + n = 17 + 64 = 81$

24. Let a circle C have its centre in the first quadrant, intersect the coordinate axes at exactly three points and cut off equal intercepts from the coordinate axes. If the length of the chord of C on the line x + y = 1 is $\sqrt{14}$, then the square of the radius of C is _____.

Sol.



$$r^2 = \left(\frac{\sqrt{14}}{2}\right)^2 + \left(r - \frac{1}{\sqrt{2}}\right)^2$$

$$r^2 = \frac{7}{2} + r^2 - \sqrt{2}r + \frac{1}{2}$$

$\sqrt{2}r = 4$

$r = 2\sqrt{2}$

$r^2 = 8$

25. If $\alpha = \int_0^{2\sqrt{3}} \log_2\left(x^2+4\right)dx + \int_2^4 \sqrt{2^x - 4}\,dx$, then α^2 is equal to _____.

Sol. As we know that

$$\int_a^b f(x)\,dx + \int_{f(a)}^{f(b)} f^{-1}(x)\,dx = bf(b) - af(a)$$

$\therefore\ \alpha = \left(2\sqrt{3}\right)(4) - (0)(2)$

$= 8\sqrt{3}$

$\Rightarrow \alpha^2 = 192$

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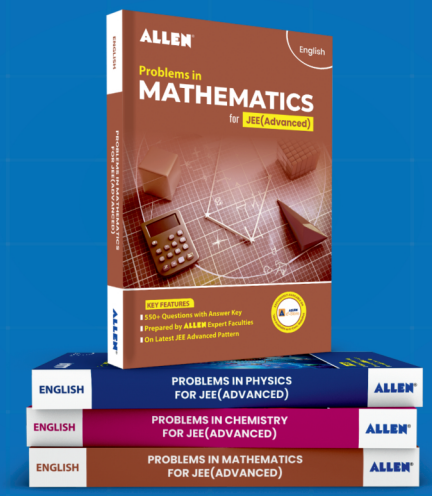


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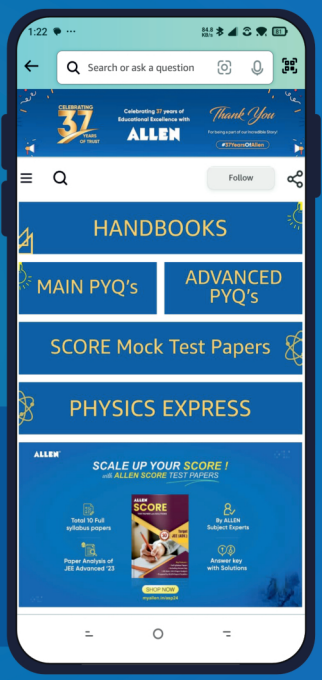
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